

Ethical Hacking Practicals

Module 1

Practical No.1: Perform Google & Whois Reconnaissance

(A). Aim: Use Google search techniques to gather information about a specific target or organization.

Theory:

Reconnaissance

Reconnaissance definition states that it is a significant instrument as a starting point of numerous data hacking and for penetration testing. The cycle includes gathering data about the target machine that could be utilized to discover its blemishes, weaknesses, and security vulnerabilities.

In the process of reconnaissance, hackers tend to be like detectives, gathering data, and information to comprehend their victims. From looking at email records to open-source data, they wish to know about the organization better than the individuals who run and look after it.

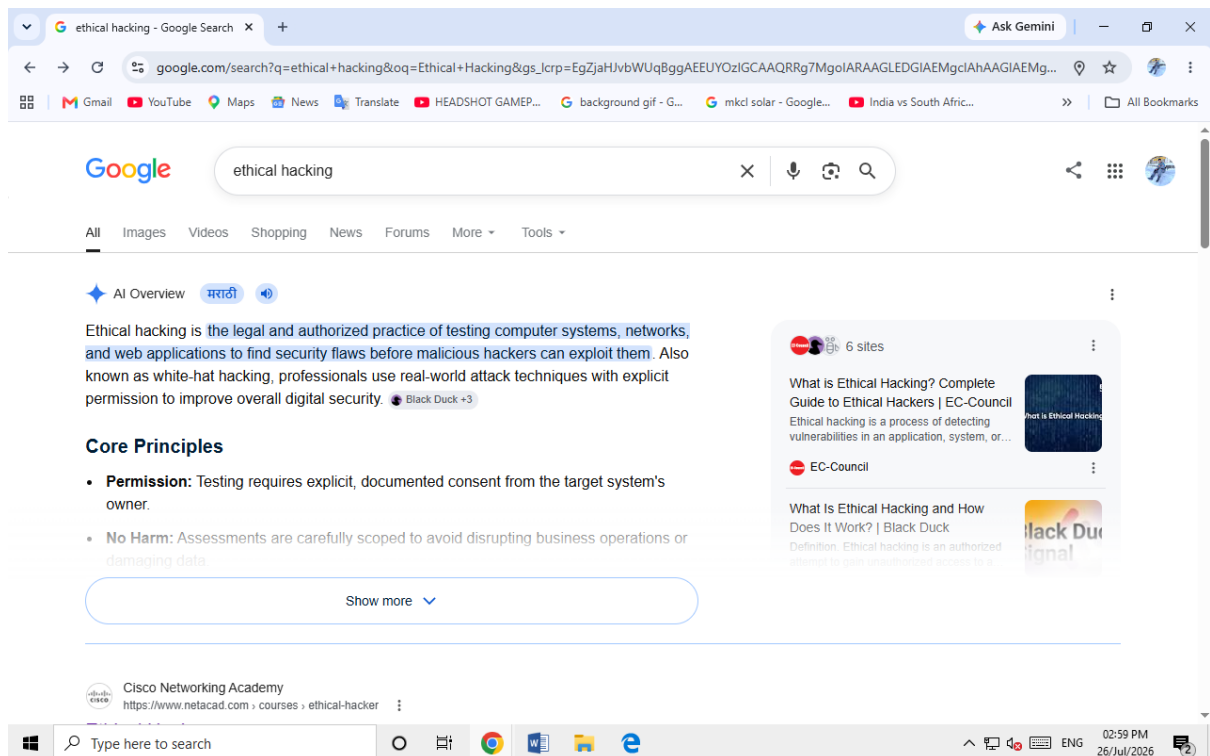
Google Reconnaissance (Using Google Search Techniques)

Google Reconnaissance, also known as **Google Hacking** or **Google Dorking**, is the process of using advanced Google search operators to find publicly available information about a target. Security professionals use these techniques during authorized penetration testing and security assessments to discover information that may unintentionally be exposed online.

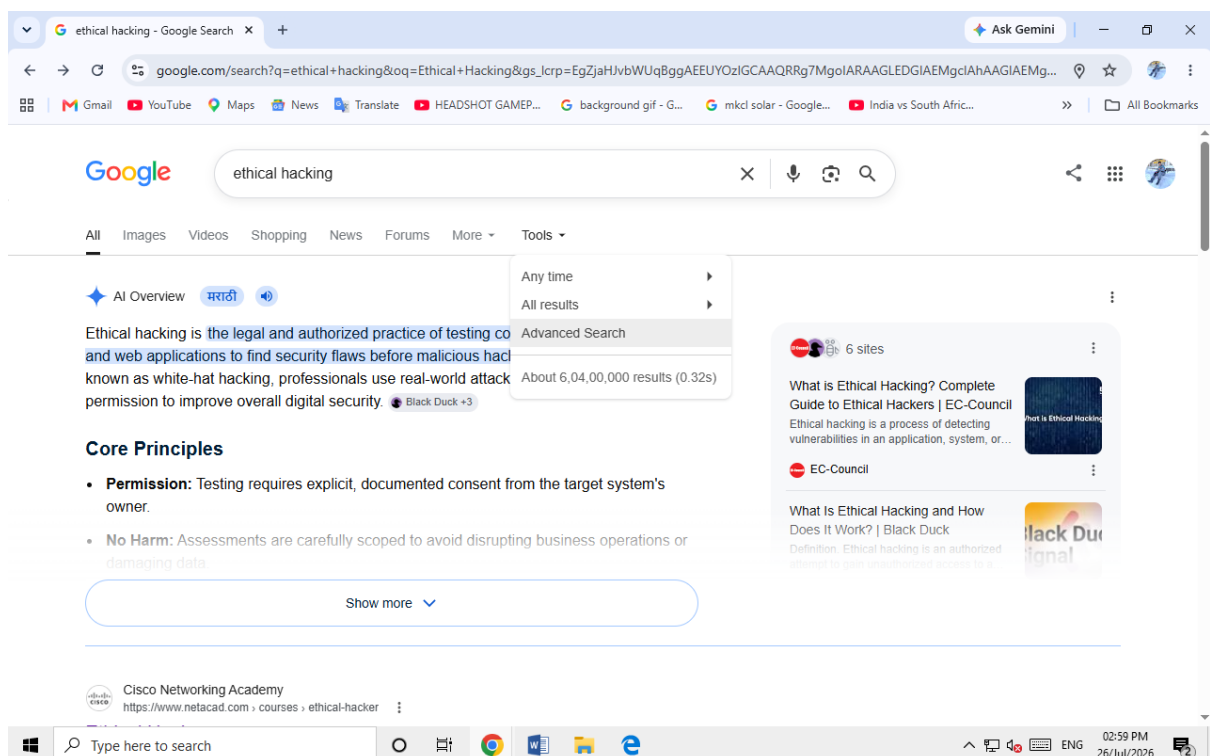
Google search operators can help locate:

- Publicly available web pages
- Company websites and documents
- Contact information
- Indexed files (such as PDFs, Word documents, and spreadsheets)
- Public login pages and other internet-facing resources.

Step 1: Open Google & Search for Ethical Hacking



Step 2: After clicking on basic search → Go to settings → Select Advanced Search → apply the filters and click on Advanced Search.



Google Advanced Search

google.com/advanced_search?q=ethical+hacking&ie=UTF-8

Google

Advanced Search

Find pages with

All of these words:

This exact word or phrase:

None of these words:

Site or domain:

You can also

- [Search pages that you've visited](#)
- [Learn how to use operators](#)
- [Customise your search settings](#)

Then narrow your results by

Type here to search

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Step 3: After that simply click on “Advanced Search” Button.

Google Advanced Search

google.com/advanced_search?q=ethical+hacking&ie=UTF-8

Site or domain:

Then narrow your results by

File type:

Last update:

Language:

Region:

Usage rights:

Advanced Search

Type here to search

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ethical hacking filetype:pdf - Google Search

google.com/search?as_q=ethical+hacking&as_epq=&as_eq=&as_sitesearch=&as_filetype=pdf&as_qdr=&lr=&cr=&tbs=&authuser=0

Google

ethical hacking filetype:pdf

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https://secl.cil.in › Ethical Hacking 5th Session PDF

Ethical Hacking

"Ethical hacking is an authorized practice of detecting vulnerabilities in an application, system, or organization's infrastructure and bypassing system. [Read more](#)

People also ask

- What exactly is ethical hacking?
- What is ethical hacker salary?
- Can we learn ethical hacking at home?
- What course do hackers study?

PVPSIT
https://www.pvpsiddhartha.ac.in › UNIT-4-CS PDF

Type here to search

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(B). Aim: Utilize advanced search operators to refine search results and access hidden information.

Theory:

Google advanced search operators are special commands and characters that filter search results. They do this by making your searches more precise and focused.

Google Search operators are broken up into three categories: “Basic”, “Advanced”, and “Unreliable”. Basic search operators are operators that modify standard text searches.

Common Search Operators

- **site:** Searches within a specific website.
Example: site:example.com
- **filetype:** Finds a specific file type such as PDF, DOCX, or XLS.
Example: site:example.com filetype:pdf
- **intitle:** Searches for pages containing a word in the title.
Example: site:example.com intitle:"login"
- **inurl:** Searches for a word in the webpage URL.
Example: site:example.com inurl:admin
- **" "** (Quotation Marks): Searches for an exact phrase.
Example: "cyber security policy"
- **-** (Minus Sign): Excludes unwanted words.
Example: network security -jobs

Uses in Ethical Hacking

- Find publicly available documents.
- Identify login portals and web pages.
- Gather information about a target organisation (OSINT).
- Detect accidentally exposed information.
- Support vulnerability assessment with permission.

Advantages

- Helps collect information quickly.
- Saves time during reconnaissance.
- Improves the accuracy of searches.
- Assists in identifying publicly exposed resources.

Search Operator	Syntax / Example	Description
" " (Quotation Marks)	"Nikola Tesla"	Searches for the exact phrase. On single words, it prevents Google from using synonyms.
OR	Tesla OR Edison	Performs a logical OR search. Results can contain either Tesla or Edison. OR must be written in uppercase.
(Pipe)	Tesla Edison	Works the same as OR. Useful as an alternative to the OR operator.
() (Parentheses)	(Tesla OR Edison) alternating current	Groups search terms and controls the order in which operators are applied.
- (Minus)	Tesla -Motors	Excludes a specific word or term from the search results.
* (Asterisk / Wildcard)	Tesla "rock * roll"	Represents any unknown word within a quoted phrase.
.. (Number Range)	Tesla announcement 2015..2017	Searches for results containing numbers within the specified range.
\$ (Dollar Sign)	Tesla deposit \$1000	Searches for prices or monetary values in US dollars.
€ (Euro Sign)	€9.99 lunch deals	Searches for prices or monetary values in Euros.
in (Unit Conversion)	250 kph in mph	Converts one unit of measurement into another directly in Google Search.

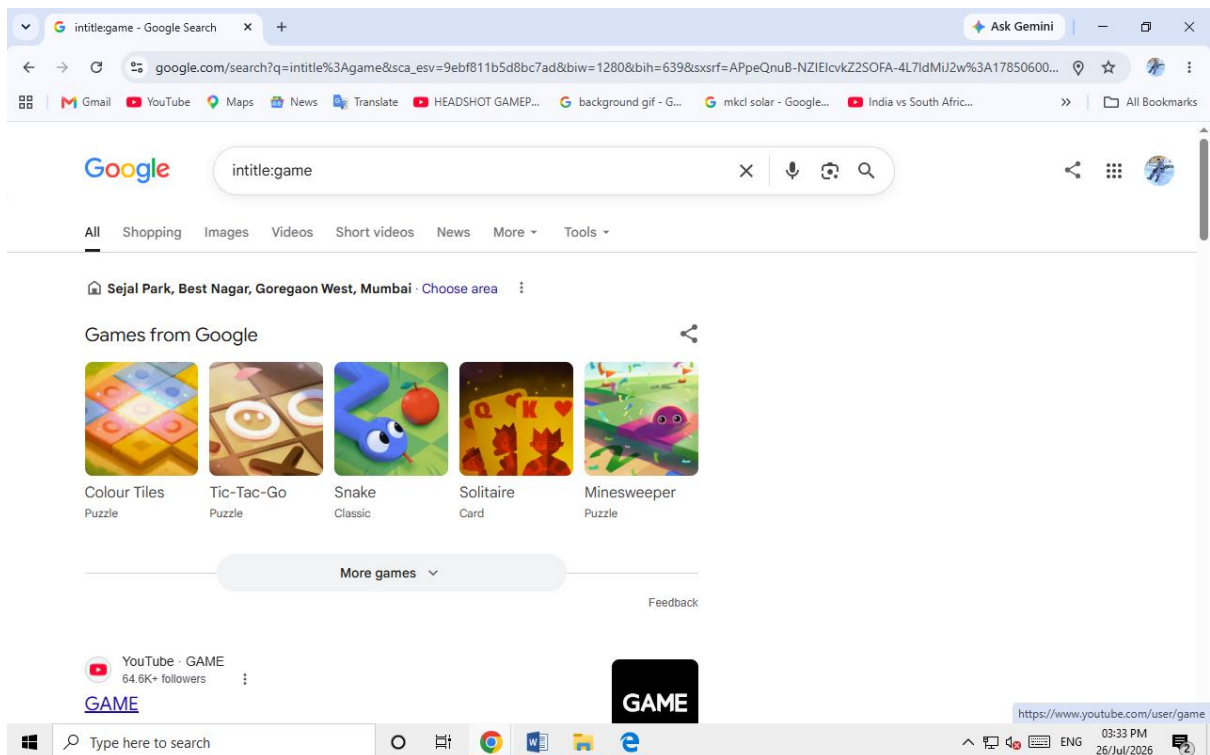
Advanced Search Operators

Advanced search operators are special commands that modify searches and may require additional parameters (such as a domain name). Advanced operators are typically used to narrow searches and drill deeper into results. The Advanced search operators are given in the below table:

S.No.	Operator	Example	Use of the Operator
1	intitle:	intitle:game	Finds web pages that contain the specified word in the page title.
2	allintitle:	allintitle:indoor game	Finds pages where all specified words appear in the page title.
3	inurl:	inurl:game	Finds pages that contain the specified word in the URL.
4	allinurl:	allinurl:indoor game	Finds pages where all specified words appear in the URL.

S.No.	Operator	Example	Use of the Operator
5	intext:	intext:game	Searches for a word or phrase only in the body text of a webpage.
6	allintext:	allintext:outdoor game	Finds pages where all specified words appear in the body text.
7	filetype:	game filetype:pdf	Restricts search results to a specific file type such as PDF, DOCX, PPT, or TXT.
8	related:	related:apple.com	Finds websites that are similar or related to the specified website.
9	AROUND(X)	apple AROUND(4) iPhone	Finds pages where two words or phrases appear w

1). Intitle



2). Allintitle

allintitle:indoor game - Google

google.com/search?q=allintitle%3Aindoor+game&sca_esv=9ebf811b5d8bc7ad&biw=1280&bih=639&txsr=APpeQnvLX80srOTpWEJxaQYRcbTRmc6Pfw...

Google allintitle:indoor game

AI Overview मराठी

An indoor game is any recreational activity, sport, or contest played entirely inside a building or home, featuring popular options like [Chess](#), [Carrom](#), and [Monopoly](#).

Classic Board and Table Games

- **Chess:** A two-player strategy game on an 8x8 grid focused on trapping the opponent's king.
- **Carrom:** A disc-flick tabletop game where players flick strikers to pocket wooden pieces.

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Decathlon Sports India
https://www.decathlon.in/blog/Best-Indoor-Games-f...
Best Indoor Games for Kids & Adults: Fun Activities at Home!
Introduction · List of the Best Indoor Games for Kids and Adults · Indoor basketball · Carrom · Darts · Archery · Tumball · Card Games. [Read more](#)

78 Best Indoor Games ideas - Pinterest
Stuck inside the house and looking for fun ideas for play time? Check out...

Indoor games and sports - Wikipedia
Indoor games and sports are a variety of structured games or competitive physical...

https://www.decathlon.in/blog/Best-Indoor-Games-for-Kids-and-Adults-Fun-Activities-at-Home

3). Inurl

inurl:game - Google Search

google.com/search?q=inurl%3Agame&sca_esv=9ebf811b5d8bc7ad&biw=1280&bih=639&txsr=APpeQnuSsuy0kv7UZww2RIUXHEjGzdEFxg%3A1785060...

Google inurl:game

All Shopping Images Videos News Forums More Tools

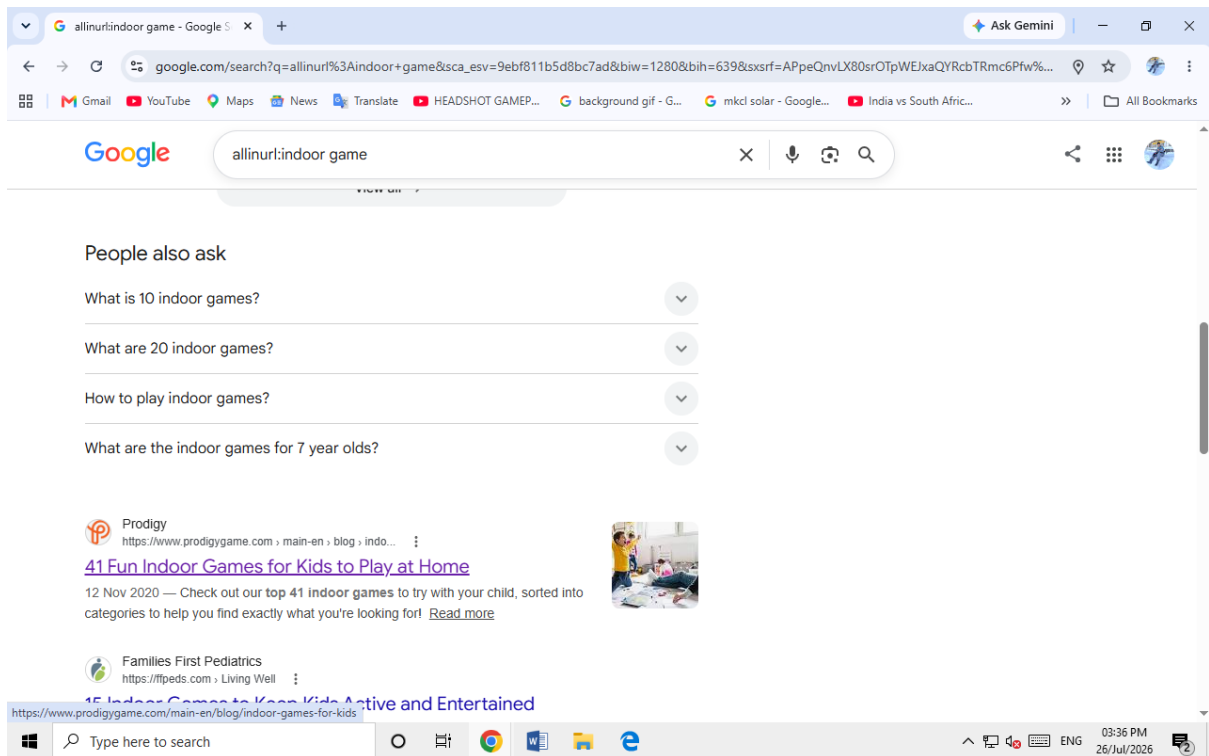
CrazyGames
https://www.crazygames.com
Free Online Games | Play Now!
Play free online games at CrazyGames, the best place to play high-quality browser games. We add new games every day. Have fun!

Alda Games
https://aldagames.com
Alda Games | Mobile games with Heart!
Alda Games is small and passionate gamedev studio. We create games that last, played by millions of gamers around the world - on mobile, PC and even consoles.

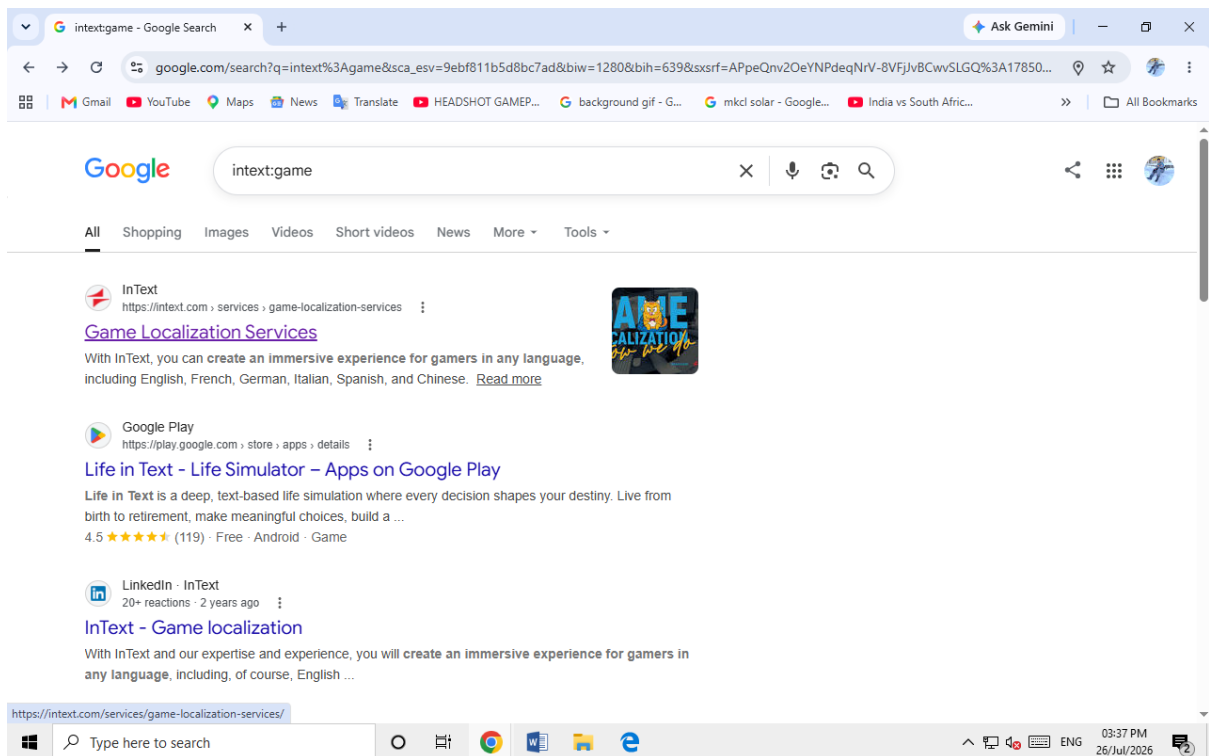
Yandex
https://yandex.com/games
Free games online to suit every taste. No downloads required
Free online games for everyone: action, puzzles, racing, two-player, and more. No downloads required! Play on your mobile phone or computer.

https://www.crazygames.com www.konoreale.com

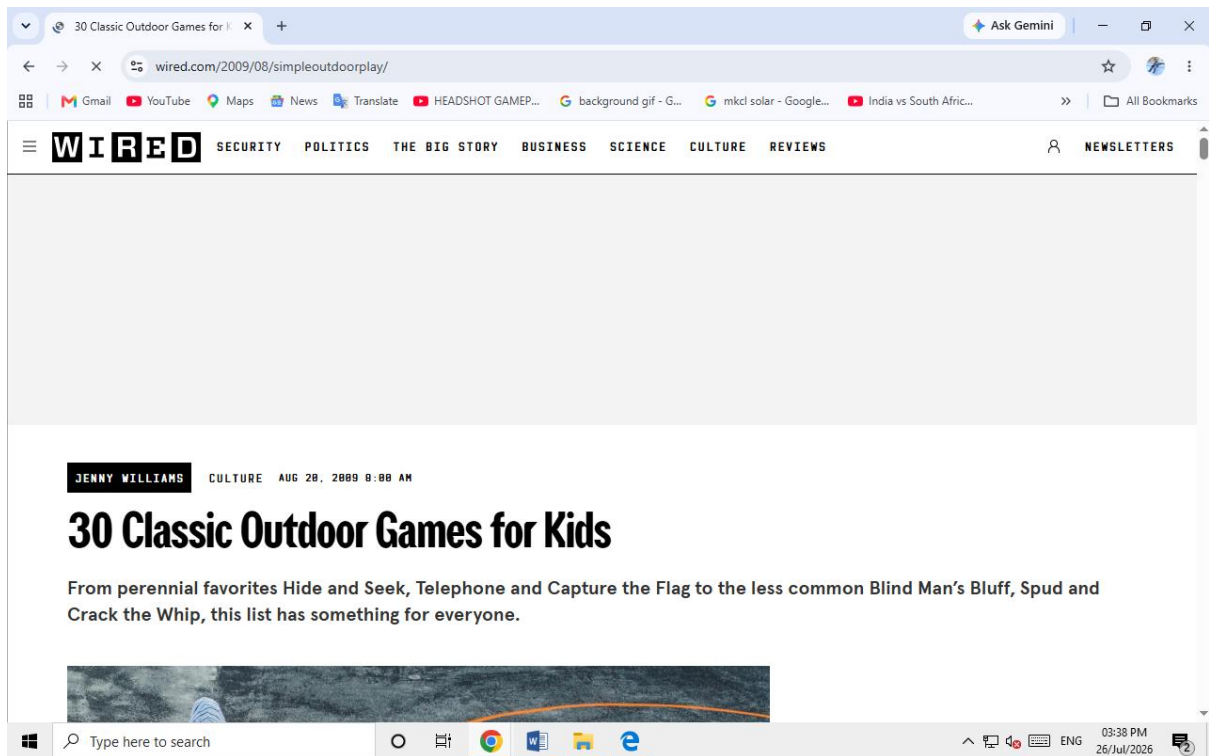
4). Allinurl



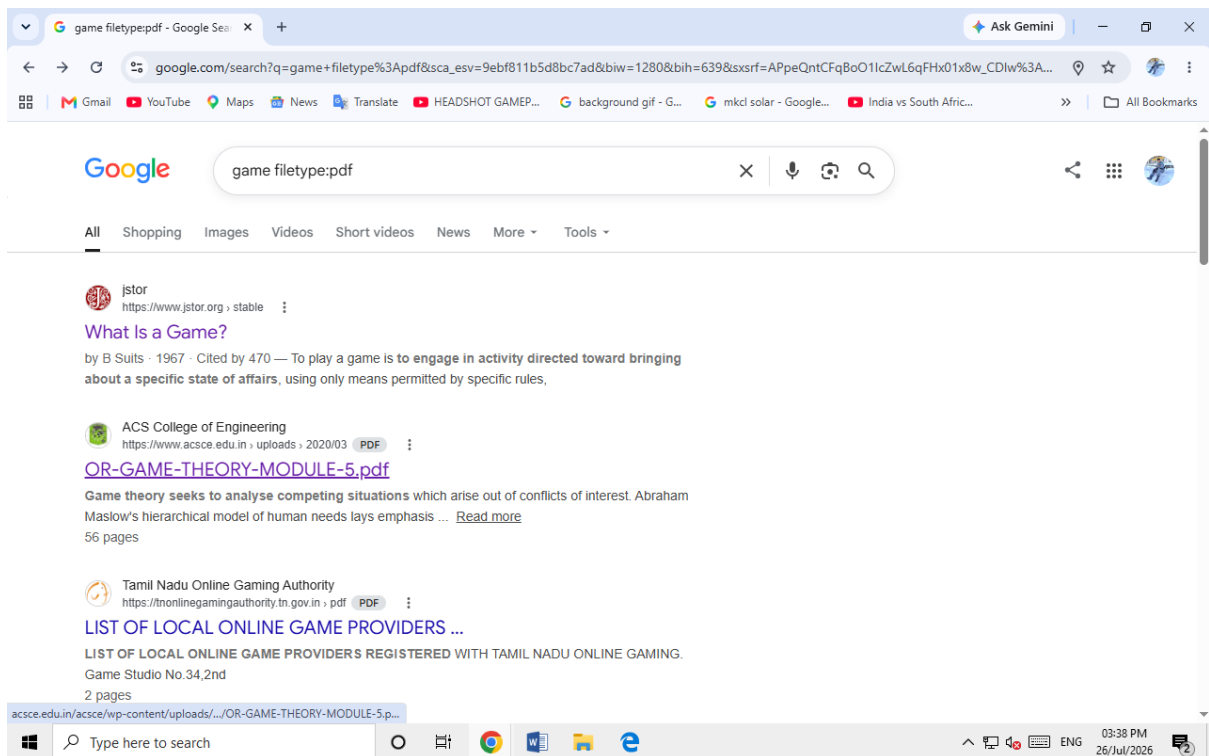
5). Intext



6). Allintext



7). Filetype



8). Related

relatedapple.com - Google Search

google.com/search?q=related%3Aapple.com&sca_esv=9ebf811b5d8bc7ad&biw=1280&bih=639&sxsrf=APpeQnt_5c39K-Vggjm-4sUA0kb4zT3yrQ%3A17...

relatedapple.com

Key Products and Services

- **iPhone:** The latest smartphone line-up featuring advanced on-device intelligence capabilities.
- **Mac:** High-performance computers powered by Apple silicon, including MacBook Pro, iMac, and Mac mini.

Support Online video, documents, forums, and direct chat with Apple Support

Show more

Apple
https://www.apple.com/...

[Apple \(India\)](#)
Discover the innovative world of Apple and shop everything iPhone, iPad, Apple Watch, Mac, and Apple TV, plus explore accessories, entertainment, ...
Missing: related:

Apple Support
https://support.apple.com/en-in/...

[Official Apple Support](#)
https://www.apple.com/in/...

9). AROUND(X)

apple AROUND(4) iPhone - Google Search

google.com/search?q=apple+AROUND%284%29+iPhone&sca_esv=9ebf811b5d8bc7ad&biw=1280&bih=639&sxsrf=APpeQnu4W5fxEZqlhJ1OCGOkJvht...

apple AROUND(4) iPhone

- **Screen:** 3.5-inch Retina display with 960 x 640 resolution.
- **Chip:** Apple A4 processor.
- **Camera:** 5-megapixel rear camera with HD video and a front selfie camera.

For a quick look at why the iPhone 4 changed phone design:

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Apple
https://www.apple.com/iphone/compare/

[Compare iPhone Models](#)
Compare features and technical specifications for iPhone 17 Pro, iPhone 17 Pro Max, iPhone Air, iPhone 17, iPhone 17e, and many more. 4K video up to 60 fps
4.5 ★ store rating (1.1K) · Shop nearby (8.1 km) · Free delivery

People also ask

What is the 4th iPhone?

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1.(C). Aim: Perform Whois lookups to retrieve domain registration information and gather details about the target's infrastructure.

Theory: Whois lookup is a technique used to retrieve publicly available information about a domain name. It provides details about the domain registration, ownership information, registrar details, registration dates, name servers, and contact information (when available).

In **ethical hacking**, Whois lookup is used during the **reconnaissance phase** to gather information about a target's internet presence with proper authorization.

Information Obtained from Whois Lookup

Information	Description
Domain Name	The registered website domain.
Registrar	The company responsible for managing the domain registration.
Registration Date	The date when the domain was created.
Expiry Date	The date when the domain registration ends.
Name Servers	Servers responsible for handling domain DNS requests.
Organization Details	Information about the domain owner or organization (if publicly available).
Contact Information	Email or contact details provided during registration (may be hidden by privacy protection).

Steps to Perform Whois Lookup

1. Open a Whois lookup website or use a Whois command-line tool.
2. Enter the target domain name.
3. Submit the query.
4. Analyze the returned registration details.

Output may include:

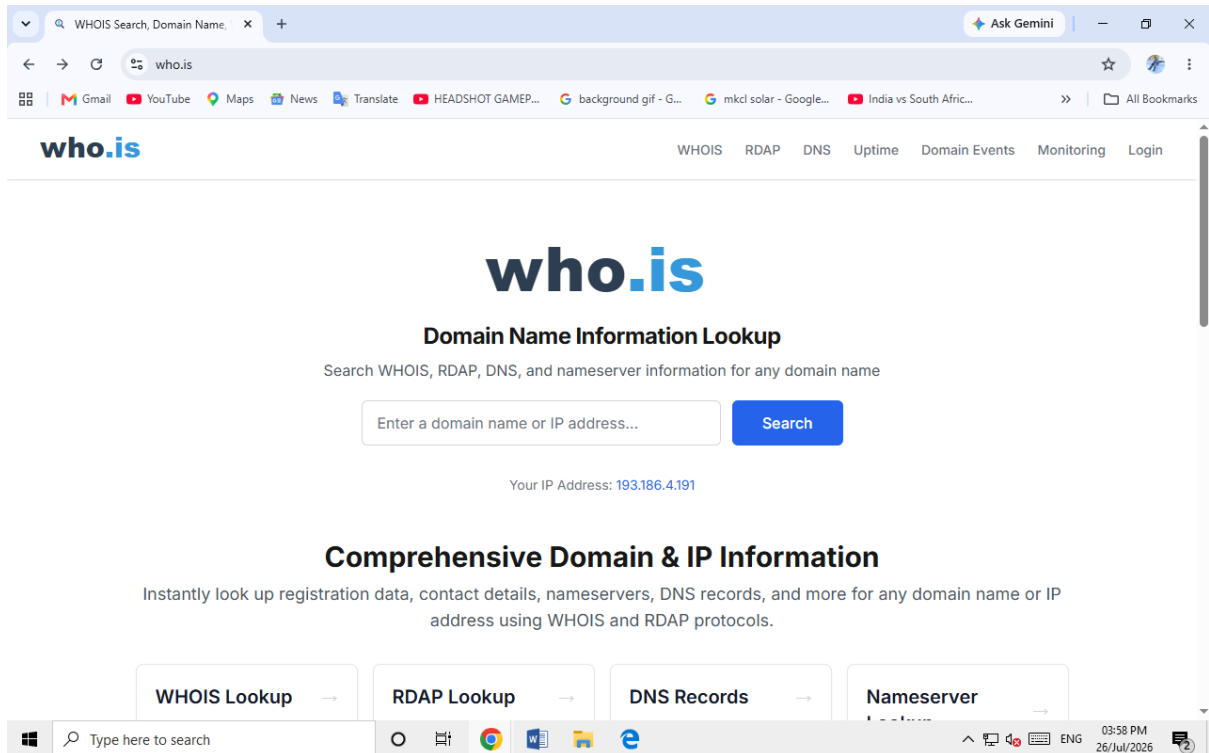
- Domain & Registrar name
- Creation and expiration dates
- Name servers
- Registration status

Uses in Ethical Hacking

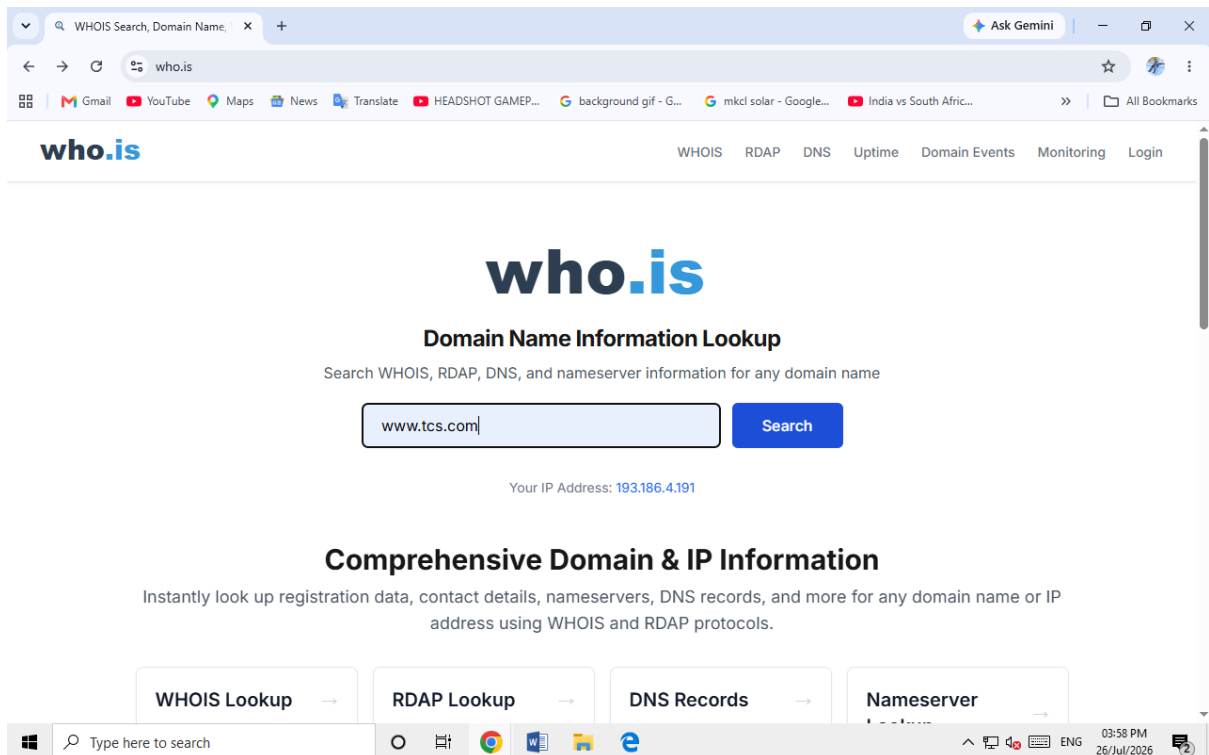
Purpose	Description
Information gathering	Collects publicly available domain details.
Domain analysis	Helps understand the target's online infrastructure.
Security assessment	Assists in identifying exposed registration information.
Network reconnaissance	Provides details useful for authorized security testing.

Practical:

Step 1: Open the “WHO.is” Website



Step 2: Enter the website name and click on “Enter button”



Step 3: Display the information about www.tcs.com

tcs.com WHOIS Domain Name | x + Ask Gemini

who.is/whois/tcs.com

Gmail YouTube Maps News Translate HEADSHOT GAMEP... background gif - G... mkl solar - Google... India vs South Afric... All Bookmarks

Whois RDAP DNS Records Certificate Uptime Diagnostics History

Hide Contact Info Refresh Data

The domain tcs.com is registered. You can still try to buy it here.

Contact Information

Abuse Contact

Phone	Email
+1.8777228662	domain [dot] operations [at] web [dot] com

Registrant Contact

Address

101 PARK AVE 26TH FL
NEW YORK, NY, 10178-0002
US

Phone	Fax
+1.9724846465	+1.9999999999

Type here to search

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tcs.com WHOIS Domain Name | x + Ask Gemini

who.is/whois/tcs.com

Gmail YouTube Maps News Translate HEADSHOT GAMEP... background gif - G... mkl solar - Google... India vs South Afric... All Bookmarks

Tech Contact

Name	Phone
Gadgil, Chetan	+1.7326923285
Email	
chetan [dot] gadgil [at] tcs [dot] com	

Registrar Information

Registrar	WHOIS Server
Network Solutions, LLC	whois.networksolutions.com
Referral URL	
http://networksolutions.com	

Important Dates

Created	Updated
November 25, 1997	October 23, 2025
Expires	
November 24, 2031	

Type here to search

ENG 04:00 PM 26/Jul/2026

tcs.com WHOIS Domain Name | x + Ask Gemini

who.is/whois/tcs.com

November 24, 2031

WHOIS data last fetched July 4, 2026

Nameservers

Hostname	IP Address
ns3.tcs.com	59.144.131.16
ns5.tcs.com	208.44.114.53
ns6.tcs.com	80.82.129.169
ns7.tcs.com	20.198.79.235
ns8.tcs.com	45.249.218.123

Domain Status

clientTransferProhibited <https://icann.org/epp#clientTransferProhibited>

Type here to search

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Step 4: Display the DNS Records

tcs.com DNS Information - Who | x + Ask Gemini

who.is/dns/tcs.com

Hosting & Network

tcs.com runs on Microsoft Azure, with email configured.

Hosting network

Microsoft Azure AS8075 Microsoft Azure

Managed DNS	Email
Self-hosted / other	Configured
IPv6 (AAAA)	CAA pinning
Not enabled	None

Sits alongside [4,999+ other sites](#) on Microsoft Azure. Stable here since we started watching (Jul 18, 2026).

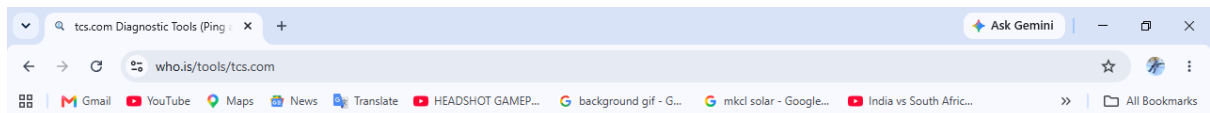
DNS Records for tcs.com

Hostname	Type	TTL	Priority	Content
tcs.com	MX	0	10	Inblrg05.tcs.com
tcs.com	MX	0	10	Inchnng02.tcs.com

Type here to search

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Step 5: Display Diagnostics



Diagnostic Tools

Run ping and traceroute diagnostics for domains and IP addresses

Search

tcs.com

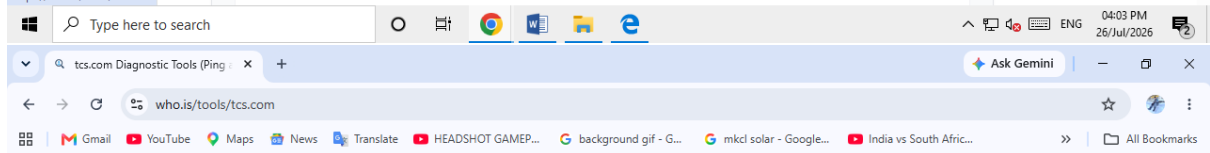
Diagnostic Tools

WhoisRDAPDNS RecordsCertificateUptimeDiagnosticsHistoryHide Contact Info

Ping Results

```
PING tcs.com (4.247.162.6): 56 data bytes
64 bytes from 4.247.162.6: seq=0 ttl=107 time=197.957 ms
64 bytes from 4.247.162.6: seq=1 ttl=107 time=195.653 ms
64 bytes from 4.247.162.6: seq=2 ttl=105 time=191.196 ms
64 bytes from 4.247.162.6: seq=3 ttl=107 time=195.680 ms
64 bytes from 4.247.162.6: seq=4 ttl=105 time=191.129 ms
```

https://who.is/tools/tcs.com



Ping Results

```
PING tcs.com (4.247.162.6): 56 data bytes
64 bytes from 4.247.162.6: seq=0 ttl=107 time=197.957 ms
64 bytes from 4.247.162.6: seq=1 ttl=107 time=195.653 ms
64 bytes from 4.247.162.6: seq=2 ttl=105 time=191.196 ms
64 bytes from 4.247.162.6: seq=3 ttl=107 time=195.680 ms
64 bytes from 4.247.162.6: seq=4 ttl=105 time=191.129 ms
```

```
--- tcs.com ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 191.129/194.323/197.957 ms
```

Traceroute Results

```
traceroute to tcs.com (4.247.162.6), 10 hops max, 46 byte packets
 1 ip-10-0-0-119.ec2.internal (10.0.0.119) 2.157 ms 1.964 ms 2.421 ms
 2 240.64.220.128 (240.64.220.128) 5.471 ms 240.64.220.129 (240.64.220.129) 5.743 ms
240.64.220.128 (240.64.220.128) 4.831 ms
 3 *
```

Cached results from: 7/26/2026, 10:28:23 AM

